

Personal Evaluation and Reflection – Sunchoeh Prince

This module has been a transformative experience, offering me the opportunity to bridge the gap between theoretical computer science concepts and real-world software engineering approaches. My primary responsibility in this project involved the end-to-end lifecycle of the machine learning model: data cleaning, pipeline development, and the model training. This journey was not just a technical exercise but a comprehensive lesson in resilience, professional development, and collaboration.

Technical Development

From a technical perspective, this project helped advance my competency in data engineering and machine learning. One of the most impactful tasks was transitioning our entire dataset from the CSV format that was given to a Parquet format instead. This was not a superficial change; it was a deliberate move towards efficiency and optimisation, which allowed faster processing and more reliable data handling. Moving my development pipeline to Kaggle was another technical milestone. By transitioning from local training to a cloud-based environment, I was able to overcome the limitations and constraints that I had previously. This allowed me to create a more robust, reproducible pipeline and allowed model training through the Kaggle API. This process required deep engagement with the model accuracy metrics. I had to learn how to iteratively refine the model, test different architectural approaches, and ensure the final output and model accuracy metrics were acceptable and met the requirements provided by Genese.

Soft Skills and Team Dynamics

Beyond the technical stack, the module was an essential crucible for my soft skills. Teamwork, communication, and collective problem-solving were the foundation of our success in this project. The synergy within our group was amazing; there was no significant conflicts, and the work environment was something to be enjoyed when we had time together. The collaborative environment made it easier to manage complex integrations such as connecting the frontend that Ethan had made to the AWS environment that Edward had been working on to connect my ML pipeline. My confidence in communication and articulating technical concepts grew significantly through our regular communication due to having to ensure that my model's logic was understandable to Ethan working on the frontend and Edward with the infrastructure.

Overcoming Challenges and Resilience

The most defining challenge of this module was personal: navigating the project and communicating with the team while recovering from surgery and an ankle injury. Being forced to work remotely while managing both physical pain and recovery and the mental toll of limited mobility was a significant hurdle. Initially, this felt like a barrier to being a present and active teammate. However, the requirement to transition to a fully remote workflow forced me to become more disciplined and proactive in my communication to ensure I was still part of the group. This period became a turning point; my contribution to the project served as a source of purpose in a way during my recovery. Overcoming both the frustration and the technical hurdles of the model pipeline proved that I could maintain productivity under duress. This experience taught me that adaptability is perhaps the most important skill for a software engineer.

Client Engagement

Dealing with our client, Genese, provided a useful insight into professional expectations. Their guidance on acceptable accuracy limits with trained models and project requirements helped ground our work. The weekly meetings were incredibly educational; they taught me how to articulate progress, log progress with the use of Trello, manage expectations, and translate feedback into actionable tasks. I also learned how to leverage project management tools like Trello effectively. Through the guidance of a mentor that was provided to us from Genese, I realized that technical development is only one half of the job; the other half is ensuring that the product aligns perfectly with the client's vision and industry standards.

Future Growth

This experience has fundamentally shifted my career trajectory. Working with the correlations between different data factors was eye-opening and sparked a genuine passion for data science. I now intend to delve deeper into machine learning, specifically focusing on how predictive modelling can solve complex, real-world problems. I plan to continue growing by exploring advanced data visualization techniques and further refining my ability to build production-ready pipelines. This module has provided the foundation, but it has also highlighted the sheer scale of the field, and I am eager to continue this pursuit of data-driven intelligence throughout my studies and into my professional career.